

Justify your answers and show your work. Unless stated otherwise, all questions worth 5pts.

Determine the radius of convergence of the following power series. Then test the endpoints to determine the interval of convergence.

1. $\sum \left(\frac{x}{3}\right)^k$

Using the root test, $\lim_{n \rightarrow \infty} \sqrt[n]{\left|\left(\frac{x}{3}\right)^n\right|} = \lim_{n \rightarrow \infty} \left|\frac{x}{3}\right| = \left|\frac{x}{3}\right| < 1$ when $-3 < x < 3$. So the radius of convergence is $(-3, 3)$.

For the endpoints, $x = -3 \Rightarrow \sum (-1)^k$ diverges, and $x = 3 \Rightarrow \sum 1$ diverges. So then the interval of convergence is $(-3, 3)$.

Alternatively note that $\sum \left(\frac{x}{3}\right)^k$ is a geometric series with ratio $\frac{x}{3}$. So it converges when $3 < x < -3$.

2. $\sum (-1)^k \frac{x^k}{5^k}$

Using the root test, $\lim_{n \rightarrow \infty} \sqrt[n]{\left|\left(\frac{-x}{5}\right)^n\right|} = \lim_{n \rightarrow \infty} \left|-\frac{x}{5}\right| = \left|-\frac{x}{5}\right| < 1$ when $-5 < x < 5$. So the radius of convergence is $(-5, 5)$.

For the endpoints, $x = -5 \Rightarrow \sum (-1)^k \frac{(-5)^k}{5^k} = \sum \frac{(5)^k}{5^k} = \sum 1$ diverges, and $x = 5 \Rightarrow \sum (-1)^k \frac{5^k}{5^k} = \sum (-1)^k$ diverges. So then the interval of convergence is $(-5, 5)$.

Alternatively note that $\sum (-1)^k \frac{x^k}{5^k} = \sum \left(-\frac{x}{5}\right)^k$ is a geometric series with ratio $-\frac{x}{5}$. So it converges when $-5 < x < 5$.

Use the geometric series $\sum_{k=0}^{\infty} x^k = \frac{1}{1-x}$, for $|x| < 1$ to find a power series representations for the following functions, and determine the interval of convergence.

3. $f(3x) = \frac{2}{1-3x}$.

Let $a = 2$ and $r = 3x$. So then $\frac{2}{1-3x} = \sum_{k=0}^{\infty} 2(3x)^k$ when $|3x| < 1$. The interval of convergence is $(-1/3, 1/3)$.

Express the following **polar** coordinates in **Cartesian** coordinates.

4. $(4, 5\pi)$

$x = 4 \cos(5\pi) = -4$ and $y = 4 \sin(5\pi) = 0$. So, $(-4, 0)$.

Express the following **Cartesian** coordinates in **polar** coordinates in at least **two** different ways.

5. $(-1, \sqrt{3})$

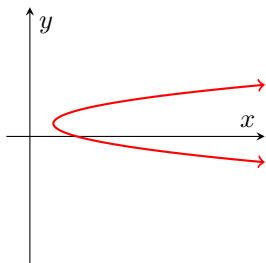
$$r^2 = (-1)^2 + (\sqrt{3})^2 = 4 \Rightarrow r = 2$$

$$\theta = \tan^{-1} \left(\frac{\sqrt{3}}{-1} \right) = -\frac{\pi}{3} + \pi = \frac{2\pi}{3} \text{ (add } \pi \text{ since the point is in quadrant II). So,}$$

$$(2, \frac{2}{3}\pi).$$

Use the parametric curve $x = t^2 + 1$ and $y = 1 - t$ for $-\infty < t < \infty$ to complete problems 6-9.

6. Sketch the parametric curve.



7. Eliminate the parameter to obtain an equation in x and y .

$$y = 1 - t \Rightarrow t = 1 - y \Rightarrow x = (1 - y)^2 + 1.$$

8. Find the slope of the tangent line when $t = 4$.

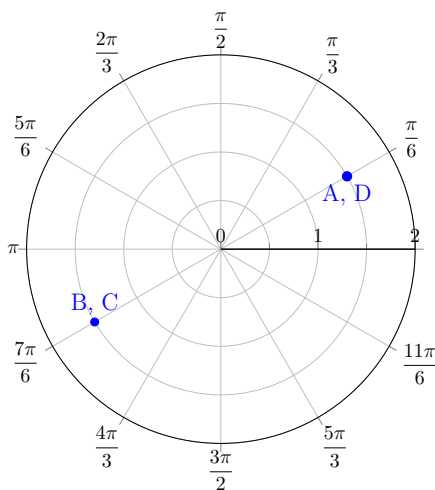
$$\frac{dy}{dx} = \frac{dy/dt}{dx/dt} = \frac{-1}{2t}. \text{ So when } t = 4 \text{ the tangent line has slope, } -\frac{1}{8}.$$

9. Find when the tangent line to the curve is a vertical line.

$$\frac{dy}{dx} = \frac{dy/dt}{dx/dt} = -\frac{1}{2t} \text{ is undefined when } 2t = 0 \Rightarrow t = 0.$$

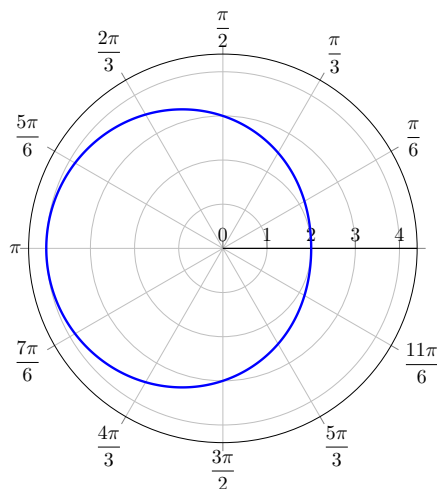
Locate the following polar coordinates on the graph.

10. A: $(1.5, \frac{\pi}{6})$, B: $(-1.5, \frac{\pi}{6})$, C: $(1.5, \frac{\pi}{6} + 3\pi)$, D: $(1.5, \frac{\pi}{6} - 8\pi)$



Use the polar equation $r = 3 - \cos \theta$ to complete problems 11-13.

11. Graph the polar equation.



12. [10 pts] Find the slope of the tangent line when $\theta = \frac{\pi}{2}$.

$$\frac{r'}{dx} = \frac{\sin \theta}{\sin \theta \cos \theta - (3 - \cos \theta) \sin \theta} \text{ at } \theta = \frac{\pi}{2} \Rightarrow \frac{1+0}{0-3} = -\frac{1}{3}$$

13. [10 pts] Find the area enclosed by the curve.

$$\begin{aligned} \frac{1}{2} \int_0^{2\pi} (3 - \cos \theta)^2 d\theta &= \frac{1}{2} \int_0^{2\pi} 9 - 6 \cos \theta + \cos^2 \theta d\theta \\ &= \frac{1}{2} \int_0^{2\pi} 9 - 6 \cos \theta + \frac{1 + \cos 2\theta}{2} d\theta \\ &= \frac{1}{2} \left[9\theta - 6 \sin \theta + \frac{1}{2}\theta + \frac{\sin 2\theta}{4} \right] \bigg|_0^{2\pi} \\ &= \frac{1}{2} [18\pi - 0 + \pi + 0] - \frac{1}{2} [0] \\ &= \frac{19}{2}\pi \end{aligned}$$